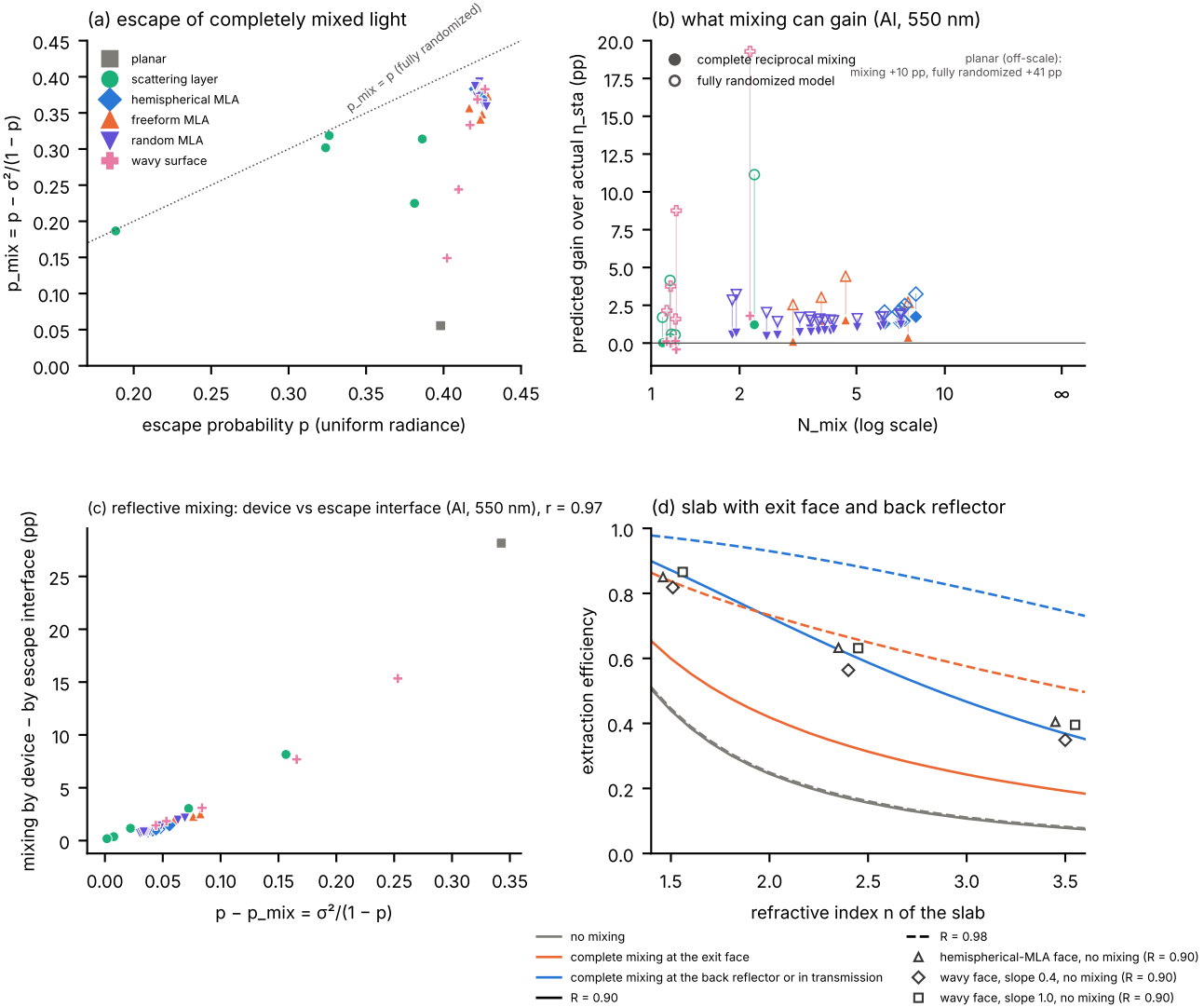
(a) escape of completely mixed light
planar
scattering layer
hemispherical MLA
freeform MLA
random MLA
wavy surface
p_mix = p (fully randomized)
p_mix = p − σ²/(1 − p)
escape probability p (uniform radiance)
(b) what mixing can gain (AI, 550 nm)
complete reciprocal mixing
fully randomized model
planar (off-scale):
mixing +10 pp, fully randomized +41 pp
predicted gain over actual η_sta (pp)
N_mix (log scale)
(c) reflective mixing: device vs escape interface (AI, 550 nm), r = 0.97
mixing by device − by escape interface (pp)
p − p_mix = σ²/(1 − p)
(d) slab with exit face and back reflector
extraction efficiency
refractive index n of the slab
no mixing
complete mixing at the exit face
complete mixing at the back reflector or in transmission
R = 0.90
R = 0.98
hemispherical-MLA face, no mixing (R = 0.90)
wavy face, slope 0.4, no mixing (R = 0.90)
wavy face, slope 1.0, no mixing (R = 0.90)